

Report

Section A

a)

Inputs: [1.2, 0.8, 2.0]

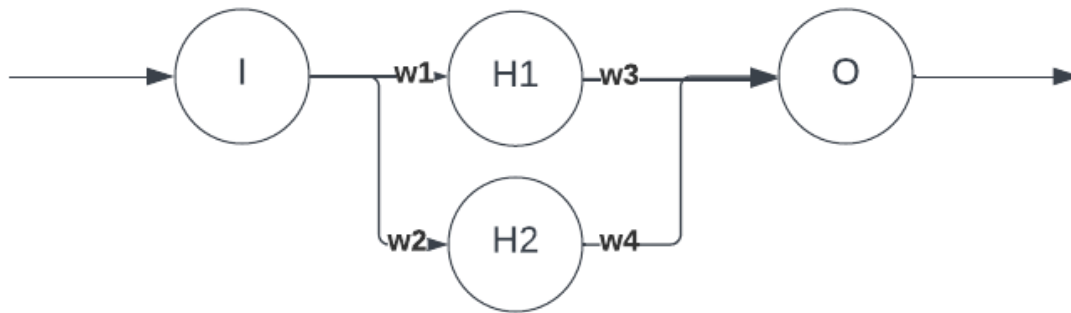
Targets: [3.0, 2.5, 4.0]

Learning rate: 0.01

Layers of network =3 (one hidden layer)

Size of layers=[1(input layer) , 2(hidden layer), 1(output layer)]

Bias=0



Forward pass:

Input=1.2

$$\text{Hidden layer output} = \text{ReLU}\left(\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) = \text{ReLU}\left(\begin{bmatrix} 0.6 \\ 0.6 \end{bmatrix}\right) = \begin{bmatrix} 0.6 \\ 0.6 \end{bmatrix}$$

$$\text{Output layer output} = 0.6 * 0.5 + 0.6 * 0.5 = 0.6$$

Input=0.8

$$\text{Hidden layer output} = \text{ReLU}\left(\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) = \text{ReLU}\left(\begin{bmatrix} 0.4 \\ 0.4 \end{bmatrix}\right) = \begin{bmatrix} 0.4 \\ 0.4 \end{bmatrix}$$

$$\text{Output layer output} = 0.4 * 0.5 + 0.4 * 0.5 = 0.4$$

Input=2

$$\text{Hidden layer output} = \text{ReLU}\left(\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) = \text{ReLU}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Output layer output= $1*0.5+1*0.5=1$

MSE = $1/3(1.2-0.6 +2-1+0.8-0.4)=0.67$

Backward pass

Output layer

Dz(output)=[0.6-1.2,0.4-0.8,1-2]=[-0.6,-0.4,-1]

Delta(output)= $1/3([-0.6,-0.4,-1].[[0.6,0.4,1],[0.6,0.4,1]])=1/3[-1.52,-1.52]=-0.5,-0.5]$

W_new(output layer)=[0.5,0.5]+0.01[0.5,0.5]=[0.55,0.55]

Hidden layer

Dz=([0.5,0.5].[-0.6,-0.4,-1])*relu_der([0.6,0.4,1])=([-0.6,-0.4,-1]. [0.5,0.5])=([[-0.3,-0.3],[-0.2,-0.2],[-0.5,-0.5]])

Dw= $1/3([-0.6,-0.4,-1]. ([[[-0.3,-0.3],[-0.2,-0.2],[-0.5,-0.5]])=$
 $1/3([0.18+0.08+0.5,0.18+0.08+0.5])=[0.253,0.253]$

db=[-1/3,-1/3]

w_new=[0.5,0.5]-[0.253,0.253]=[0.247,0.247]

b) For test sample we use the following expression:

$$\begin{aligned}w^T x + b &= \left(\sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} \right)^T x + b \\&= \sum_{i=1}^m \alpha_i y^{(i)} \langle x^{(i)}, x \rangle + b \\&= \sum_{i=1}^m \alpha_i y^{(i)} K(x_{test}, x_i) + b\end{aligned}$$

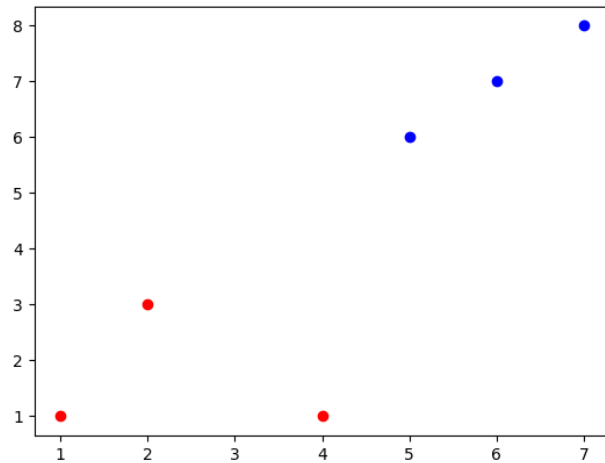
For gaussian kernel:

$$K(x, z) = \exp \left(-\frac{\|x - z\|^2}{2\sigma^2} \right)$$

final classification expression:

$$f(x_{test}) = \text{sign} \left(\sum_{i=1}^n \alpha_i y^{(i)} \exp \left(-\frac{\|x_{test} - x_i\|^2}{2\sigma^2} \right) + \gamma \right) \quad \text{where } \gamma = \text{bias term}$$

c)



- a) Yes the points are linearly separable as seen from the above picture
b) For the equation of the decision boundary we solve the following equations:

By inspection we have 3 support vectors:

$$\begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

Inserting a bias into the vectors

$$\begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

And now solving the following equations:

$$\alpha_1 \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix} \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} = +1$$

$$\alpha_1 \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = +1$$

$$18\alpha_1 + 27\alpha_2 + 12\alpha_3 = 1$$

$$27\alpha_1 + 62\alpha_2 + 29\alpha_3 = -1$$

$$12\alpha_1 + 29\alpha_2 + 14\alpha_3 = 1$$

On solving these questions:

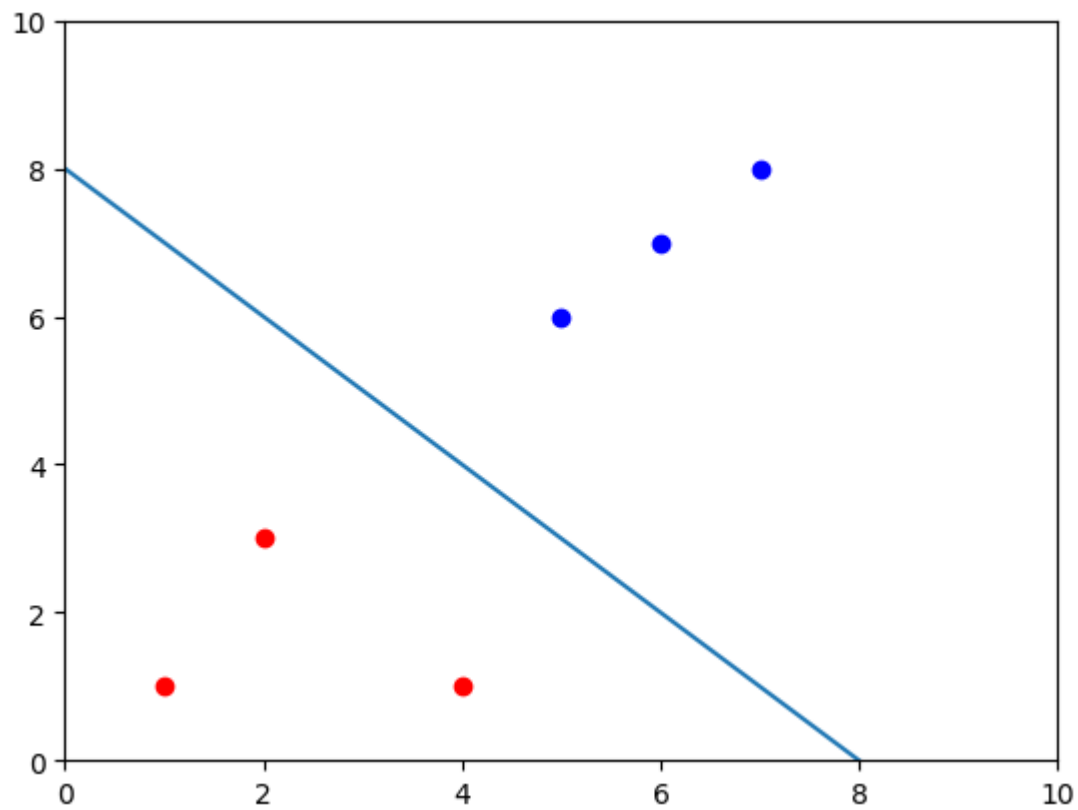
$$\alpha_1 = \frac{2}{3} \quad \alpha_2 = -\frac{7}{3} \quad \alpha_3 = \frac{13}{3}$$

$$w = \sum \alpha_i s_i$$

$$= \frac{2}{3} \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} - \frac{7}{3} \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix} + \frac{13}{3} \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1/3 \\ -1/3 \\ 8/3 \end{pmatrix}$$

Equation of hyperplane-> $y+x=8$



c)

support vectors

distance from planes

$$\begin{aligned}
 (2,3) &\rightarrow \frac{3}{\sqrt{2}} \\
 (6,7) &\rightarrow \frac{7}{\sqrt{2}} \\
 (5,6) &\rightarrow \frac{6}{\sqrt{2}} \\
 (4,1) &\rightarrow \frac{1}{\sqrt{2}} \\
 (1,1) &\rightarrow \frac{1}{\sqrt{2}} \\
 (7,8) &\rightarrow \frac{8}{\sqrt{2}}
 \end{aligned}$$

Since nearest distances are (2,3) (4,1) (5,6) they are the support vectors

$$\begin{aligned}
 \text{margin} &= \frac{1}{\|w\|} \\
 &= \frac{1}{\sqrt{1^2 + 1^2}} = \frac{1}{\sqrt{2}} = 0.707
 \end{aligned}$$

- d) If we remove any of the support vectors the decision boundary would either shift or rotate.
If we remove (4,1) or (5,6) the boundary would shift but if we remove (2,3) it will rotate.

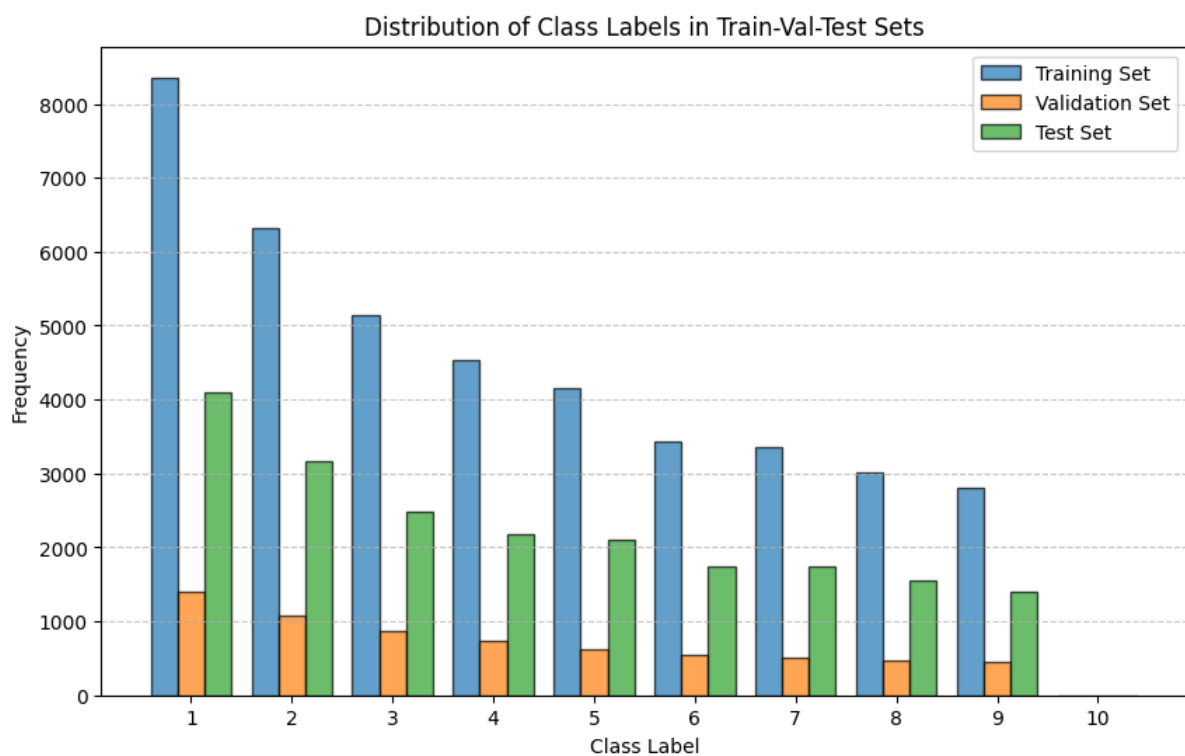
Section B

Graphs and code ipynb

Section C

1)

iii)



iv)



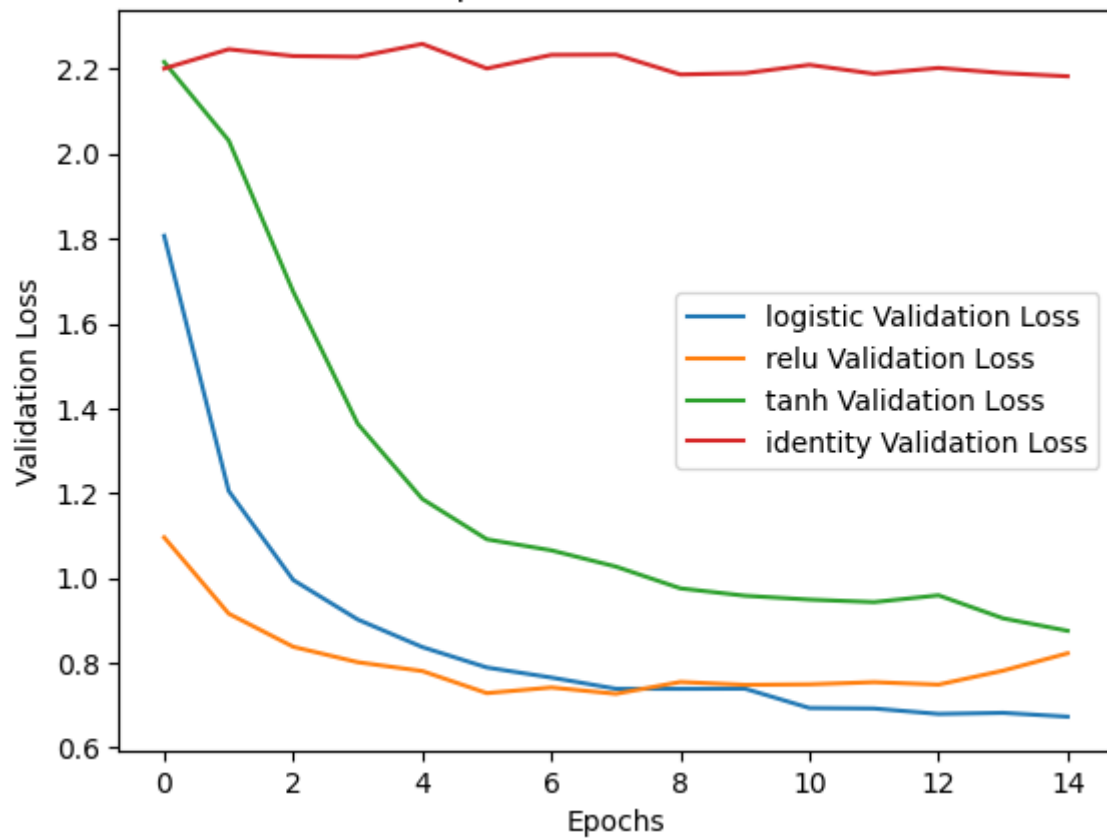
2)

ii) optimal parameters: batch size=150, hidden layer=(300,300) epoch=15

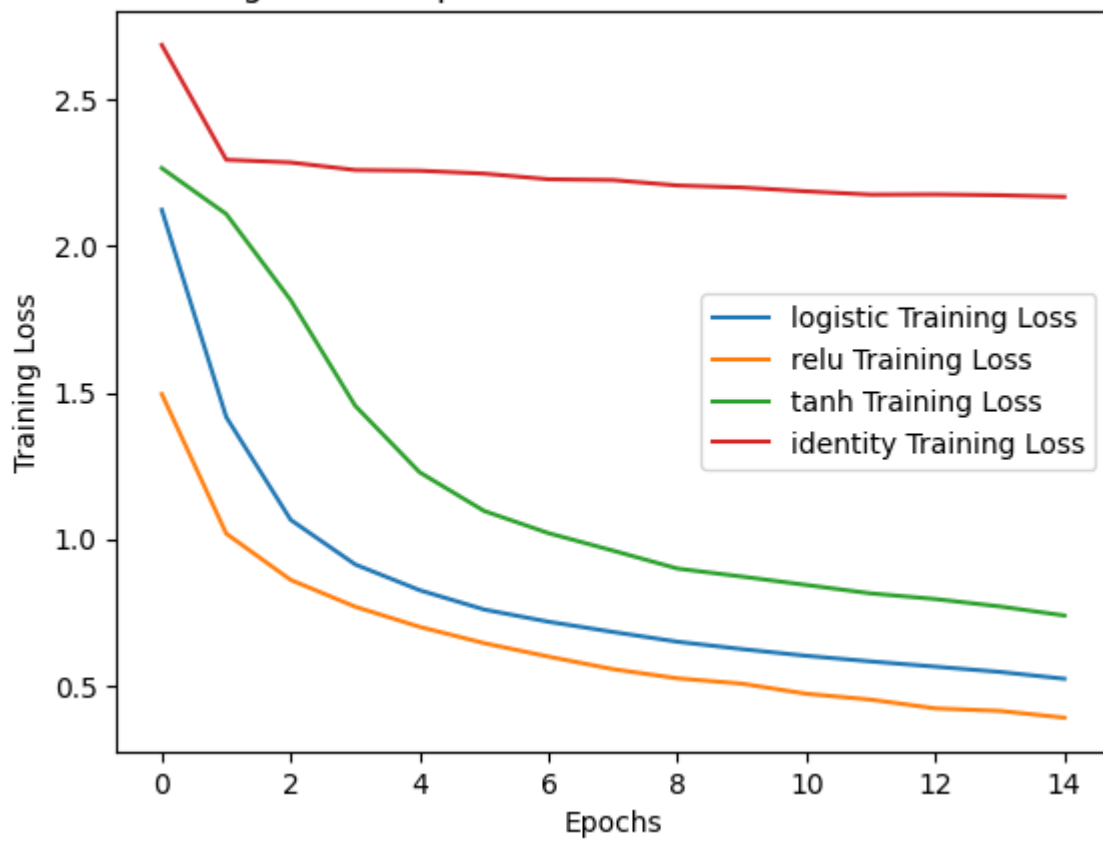
iii) losses in ipynb

iv)

Validation Loss vs. Epochs for Different Activation Functions



Training Loss vs. Epochs for Different Activation Functions



v) The model can effectively learn with sigmoid, relu, and tanh activation function with the best learning with relu activation. This is because non linear functions enable the network to approximate complex functions and incorporate non linear relationship between data.

The model performs poorly with linear activation because then they act as linear models like linear regression or classification. The image data is not linearly separable which causes the linear activation to perform poorly.

vi)

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Train Accuracy: 0.8886369305426427
Test Accuracy: 0.8007553007553008
Val Accuracy: 0.7923398328690808
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This is the best accuracy with batch size=150, hidden layer=(300,300), activation=relu and iterations=15.

These accuracy was obtained by using multiple gridsearch cv on various parameters and trying the activation functions.

3)

i) Images in ipynb

ii) Possible reasons for misclassification:

Distrations present in image: More than one number is present

Blurry image: The image is too blurry

Insufficient training data for proper classification